

Automated Analyst Report

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Query: Analyse supplier performance across all product types. Identify which product categories have the highest defect rates, which shipping carriers offer the best revenue-to-cost ratio, flag any SKUs with failed inspection results, and highlight stockout risk where lead times are long relative to stock levels. report findings must be backed by solid proof from within the data

****EXECUTIVE SUMMARY**** Hair care products exhibit the highest average defect rate (2.48 %) while Carrier C delivers the strongest revenue to shipping cost efficiency (US\$1,138.56 per shipping dollar). Thirty six SKUs failed inspection, with several high defect, low stock items (e.g., SKU 42, SKU 65) and a handful of SKUs (e.g., SKU 34, SKU 68) showing extreme stockout risk due to minimal inventory and long lead times.

****KEY FINDINGS**** - ****Defect concentration by product type:**** Hair care = 2.483 % (highest), Skincare = 2.335 %, Cosmetics = 1.919 % (lowest). - ****Carrier efficiency:**** Carrier C achieves the best revenue per shipping dollar ratio (US\$1,138.56) and a low avg. defect rate (2.60 %); Carrier B follows with US\$1,055.70 per shipping dollar and the lowest carrier level defect rate (1.76 %). - ****Failed inspection SKUs:**** 36 SKUs flagged; the top three by defect rate are SKU 42 (4.94 %, stock 22), SKU 65 (4.91 %, stock 57) and SKU 50 (4.75 %, stock 31). All belong to Skincare or Cosmetics and are shipped primarily via Carrier C. - ****Stockout risk hotspots:**** SKU 34 (Skincare) scores 13.0 (stock 1, lead time 26 days); SKU 68 (Hair care) scores 8.0 (stock 0, lead time 8 days); SKU 24 (Hair care) scores 5.2 (stock 4, lead time 26 days). These three alone represent the top 30 % of risk scores. - ****Margin resilience:**** Gross margin ratios remain above 99 % for all categories (Cosmetics 0.993, Hair care 0.991, Skincare 0.992), indicating pricing strength despite defect and logistics challenges. - ****Transport mode impact:**** Air shipments have the lowest average defect rate (1.824 %) across 26 SKUs, while Road shows the highest (2.621 %); however, the majority of high defect SKUs are still routed via Road and Rail.

****RISKS AND RECOMMENDATIONS**** The convergence of high defect rates in hair care, concentration of failed inspection SKUs on Carrier C, and acute stockout risk for low inventory SKUs (e.g., SKU 34, SKU 68) threatens service levels and margin protection. Prioritize a quality improvement program for hair care suppliers, renegotiate carrier contracts to shift high volume, high defect shipments from Carrier C to the more efficient Carrier B, and implement safety stock buffers for the top risk SKUs to mitigate lead time exposure.